

Name: _____

Reg No: _____

Total Marks: 10

1. (3 points) (a) What is the fundamental limitation of standard autoencoders that VAEs address? Briefly explain

2. (3 points) For a VAE with latent dimension $d = 2$, the encoder outputs:

$$\mu = [0.5, -0.3], \quad \log \sigma^2 = [-1.0, 0.5]$$

Noise samples: $\epsilon_1 = 0.2$, $\epsilon_2 = -0.5$ (sampled from $\mathcal{N}(0, 1)$).

- (a) Using the reparameterization trick, compute the latent code z .

3. (4 points) State whether the following statements are **True** or **False**:

- (a) (1 point) The reparameterization trick makes random sampling differentiable.
- (b) (1 point) After training a VAE, we discard the decoder and keep only the encoder for generation.
- (c) (1 point) KL divergence in VAEs acts as a regularizer that pushes $q(z|x)$ towards $\mathcal{N}(0, I)$.
- (d) (1 point) A VAE with a very large KL weight will produce sharp, diverse generations.

Answer Key

- 1(a) Standard autoencoders cannot generate new data because their latent space is sparse/discontinuous (points only exist where training data mapped). VAEs create a regularized, continuous latent space.
- 1(b) Mean (μ) and variance (σ^2) or log variance ($\log \sigma^2$).
- 2(a) $\sigma_1^2 = e^{-1.0} = 0.368$, $\sigma_1 = \sqrt{0.368} = 0.607$; $\sigma_2^2 = e^{0.5} = 1.649$, $\sigma_2 = \sqrt{1.649} = 1.284$
- 2(b) $z_1 = 0.5 + 0.607 \times 0.2 = 0.5 + 0.121 = 0.621$; $z_2 = -0.3 + 1.284 \times (-0.5) = -0.3 - 0.642 = -0.942$
- 3(a) $D_{KL} = -0.5(1 + \log(0.5) - 0.5^2 - 0.5) = -0.5(1 - 0.693 - 0.25 - 0.5) = -0.5(1 - 1.443) = 0.222$
- 3(b) $D_{KL} = -0.5(1 + \log(1.649) - (-0.3)^2 - 1.649) = -0.5(1 + 0.5 - 0.09 - 1.649) = -0.5(1.5 - 1.739) = -0.5(-0.239) = 0.120$
- 4(a) True
- 4(b) False (we discard the encoder, keep decoder for generation)
- 4(c) True
- 4(d) False (large KL weight forces $q(z|x) \approx \mathcal{N}(0, 1)$, causing information loss and blurry outputs)